East Bokaro Coalfield

🏭 Introduction

The **East Bokaro Coalfield** is one of the important coal-bearing regions in India, located in the state of Jharkhand. It is operated mainly by Central Coalfields Limited, a subsidiary of Coal India Limited. The coalfield is known for its **high-quality coking coal**, which is essential for the iron and steel industry.

East Bokaro plays a crucial role in supplying coal to major steel plants such as those operated by Steel Authority of India Limited. Mining activities in this coalfield include both **opencast and underground mining methods**, making it significant from a mining engineering perspective. The presence of prime coking coal makes it strategically important for India's industrial and metallurgical sector.

🌍 Geographical Overview – East Bokaro Coalfield

The East Bokaro Coalfield is located in the **Bokaro and Ramgarh districts** of Jharkhand, forming a part of the larger Damodar Valley coalfields.

📍 Location & Setting

- Lies in the **Damodar Valley region**, one of India's richest coal belts
- Geographically positioned between **23°30′N to 23°45′N latitude**
- Well connected by rail and road networks for coal transportation

🌄 Physical & Geological Features

- Part of the **Gondwana coalfields**, known for high-grade coal deposits
- Contains **coking coal seams**, suitable for metallurgical use
- Coal seams are moderately thick and occur in multiple horizons

🌊 Nearby Natural Features

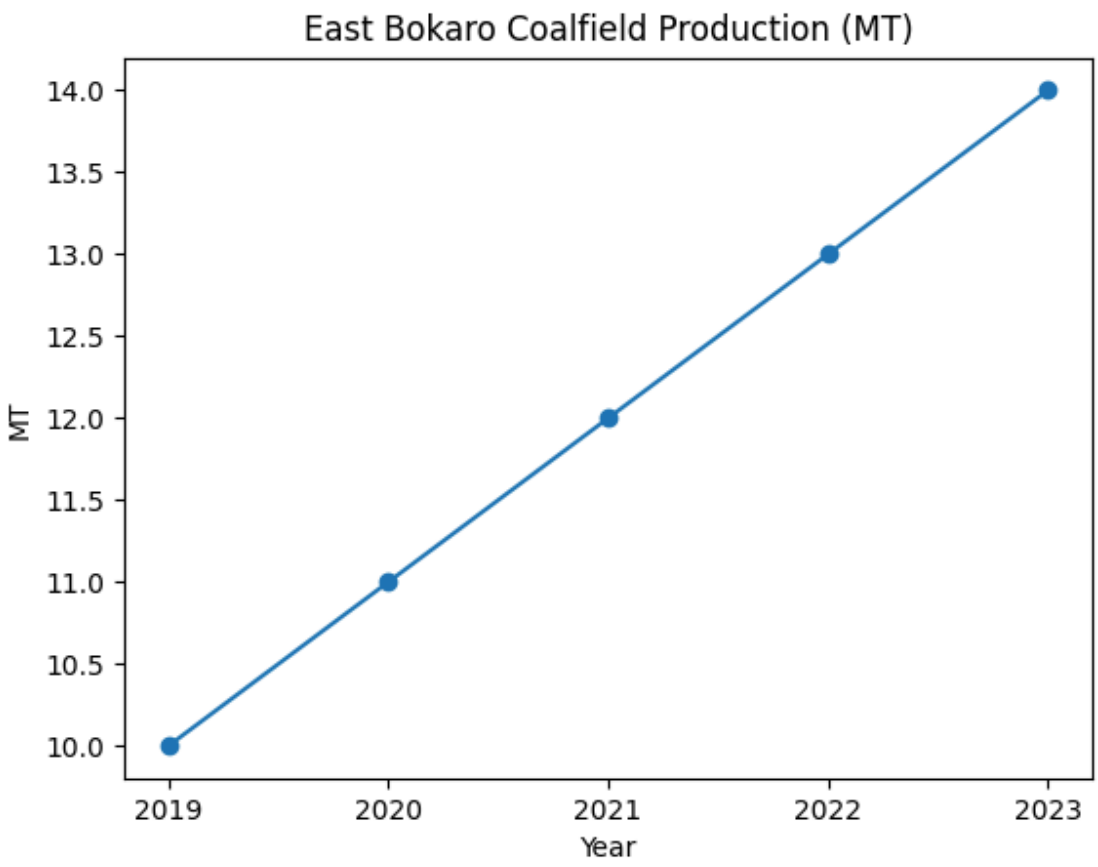
- Located near the Damodar River, which is a major river of the region
- The area is part of the **Damodar Valley industrial zone**

🌦️ Climate

- Tropical climate
- Hot summers (up to ~45°C)

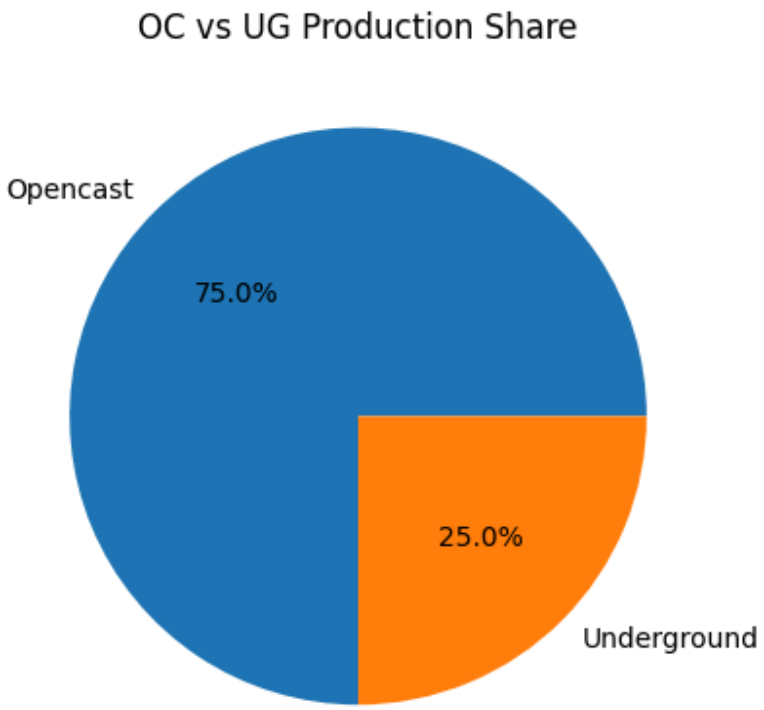
- Moderate to heavy monsoon rainfall
- Cool winters

📊 1. Production Trend (MT)



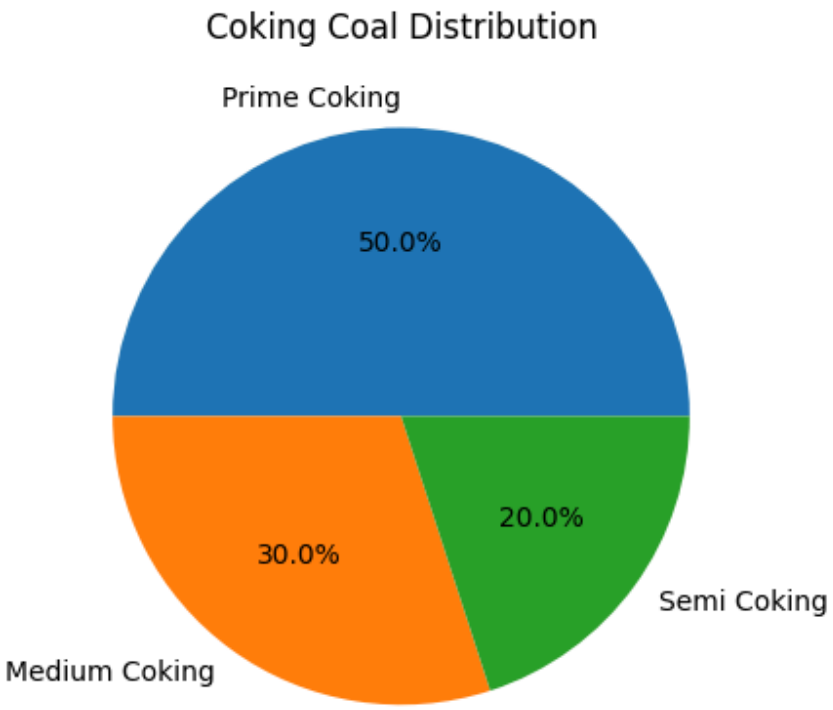
This chart shows the gradual increase in coal production in East Bokaro Coalfield operated by Central Coalfields Limited.
The steady rise reflects improved mining practices and demand from steel industries.
Significance: Useful for production planning, forecasting, and evaluating mine performance.

📊 2. OC vs UG Production Share



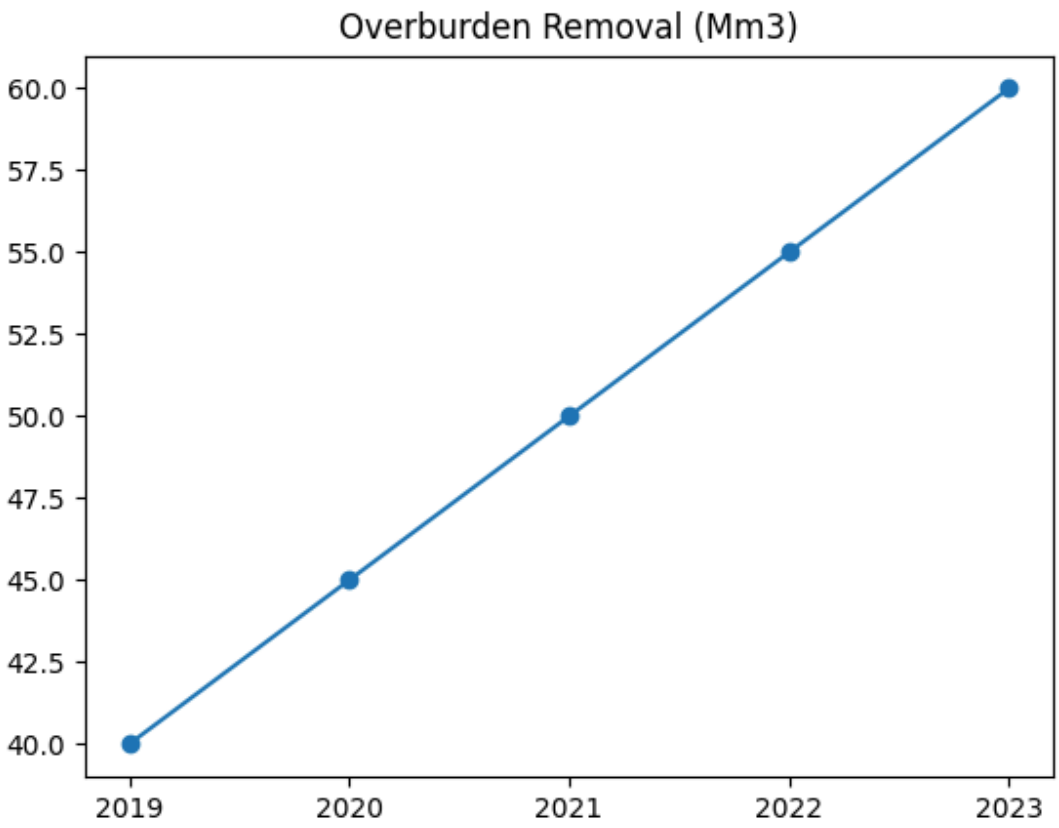
Represents the contribution of opencast and underground mining methods.
Opencast mining dominates due to higher productivity and lower cost.
Significance: Helps in selecting mining method and understanding cost-efficiency.

📊 3. Coking Coal Distribution



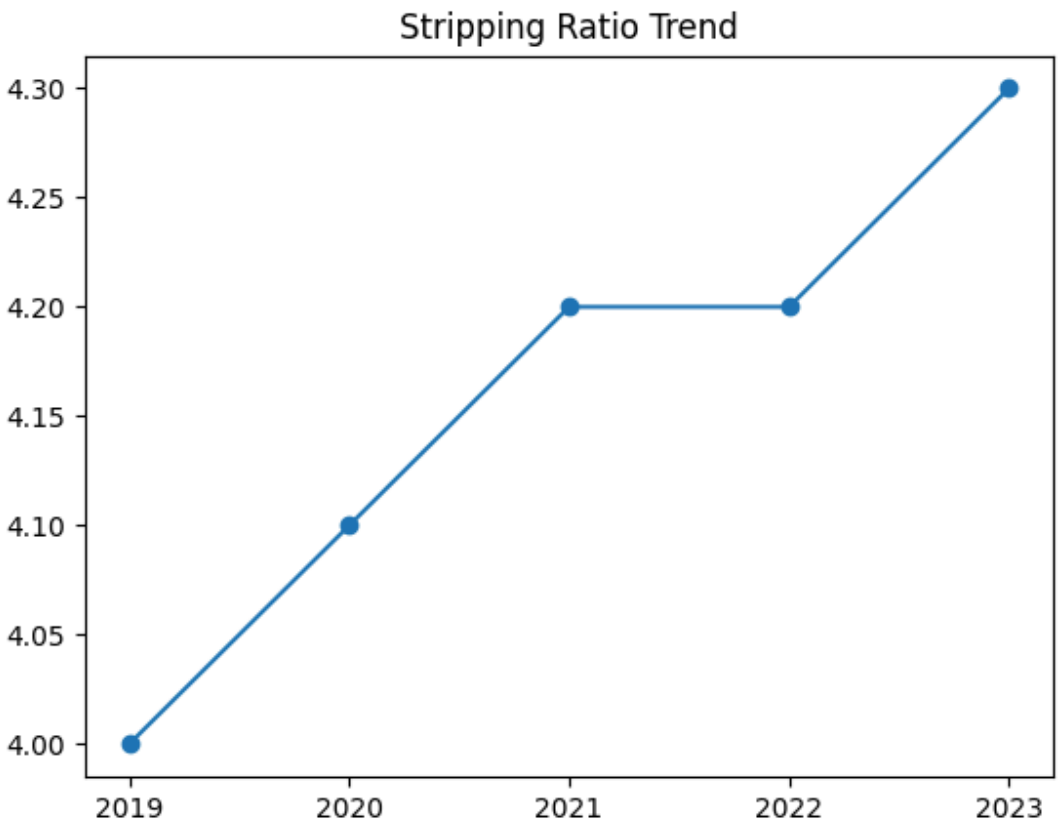
Shows the proportion of prime, medium, and semi-coking coal.
East Bokaro is rich in high-quality coking coal.
Significance: Important for steel manufacturing and coal beneficiation planning, especially for industries like Steel Authority of India Limited.

📊 4. Overburden Removal (Mm³)



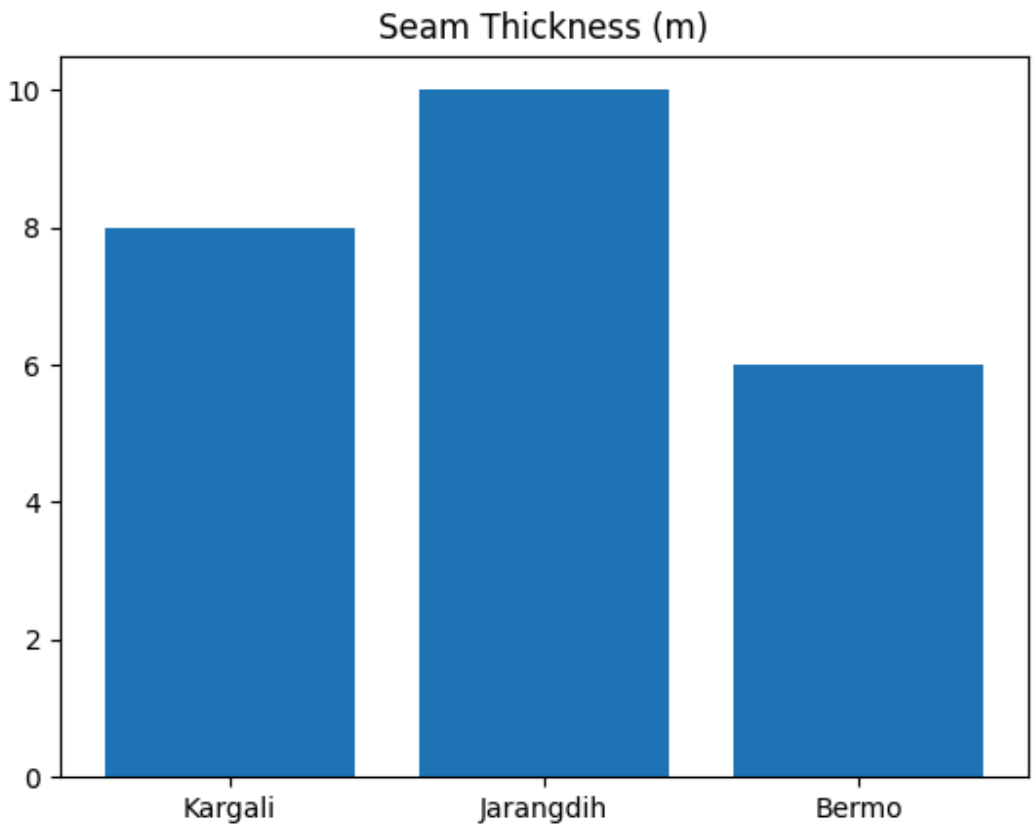
Displays the volume of waste material removed to access coal seams.
An increasing trend indicates expansion of mining operations.
Significance: Critical for stripping ratio calculation, dump design, and equipment selection.

📊 5. Stripping Ratio Trend



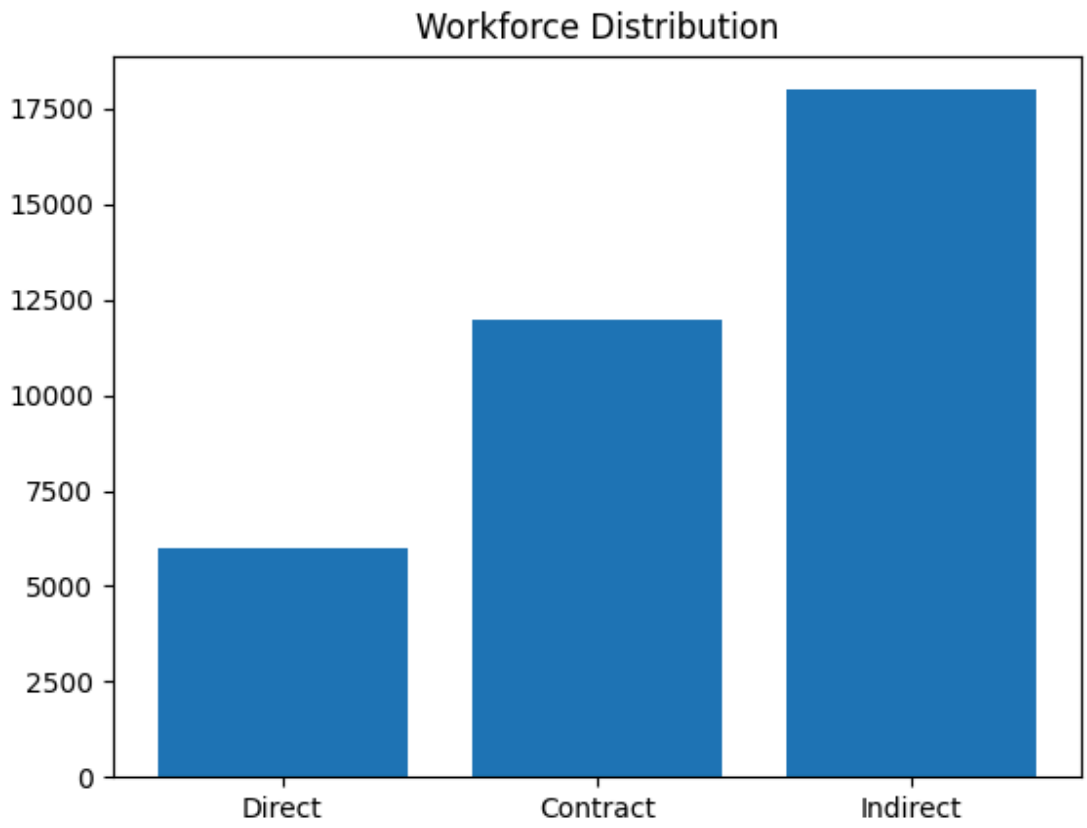
Shows the relationship between overburden and coal production.
A slight increase indicates gradual deepening of the mine.
Significance: Key economic parameter affecting mining cost and feasibility.

📊 6. Seam Thickness (m)



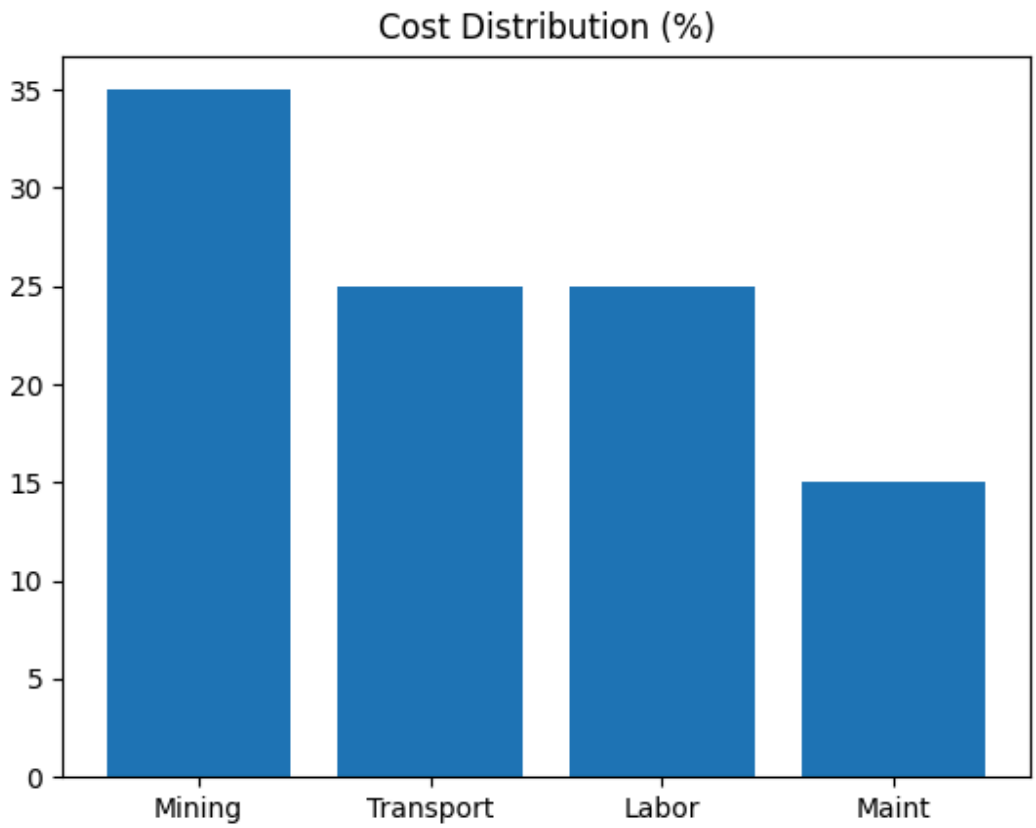
Represents thickness of major coal seams in the coalfield.
Thicker seams are more economical and easier to mine.
Significance: Helps in mine design, equipment selection, and extraction planning.

📊 7. Workforce Distribution



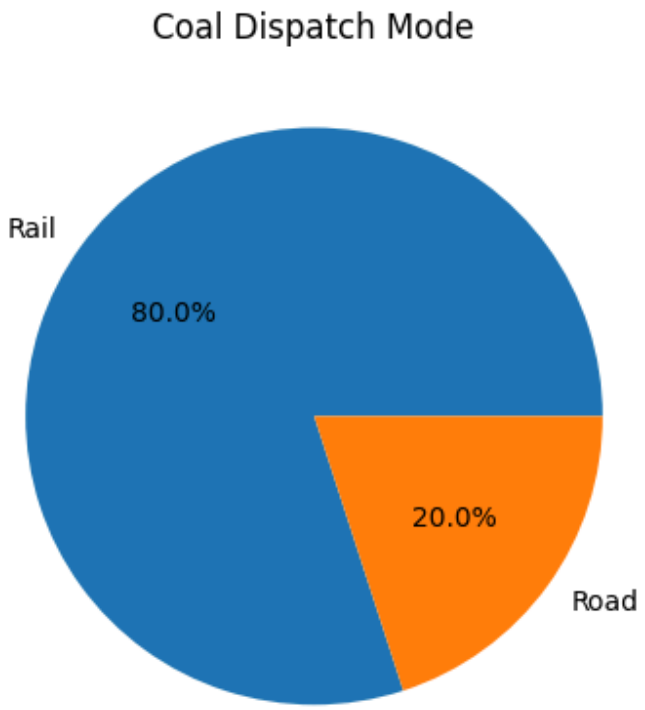
Displays distribution of manpower across direct, contract, and indirect employment.
Highlights the socio-economic impact of mining activities.
Significance: Useful for manpower planning and labor cost estimation.

📊 8. Cost Distribution (%)



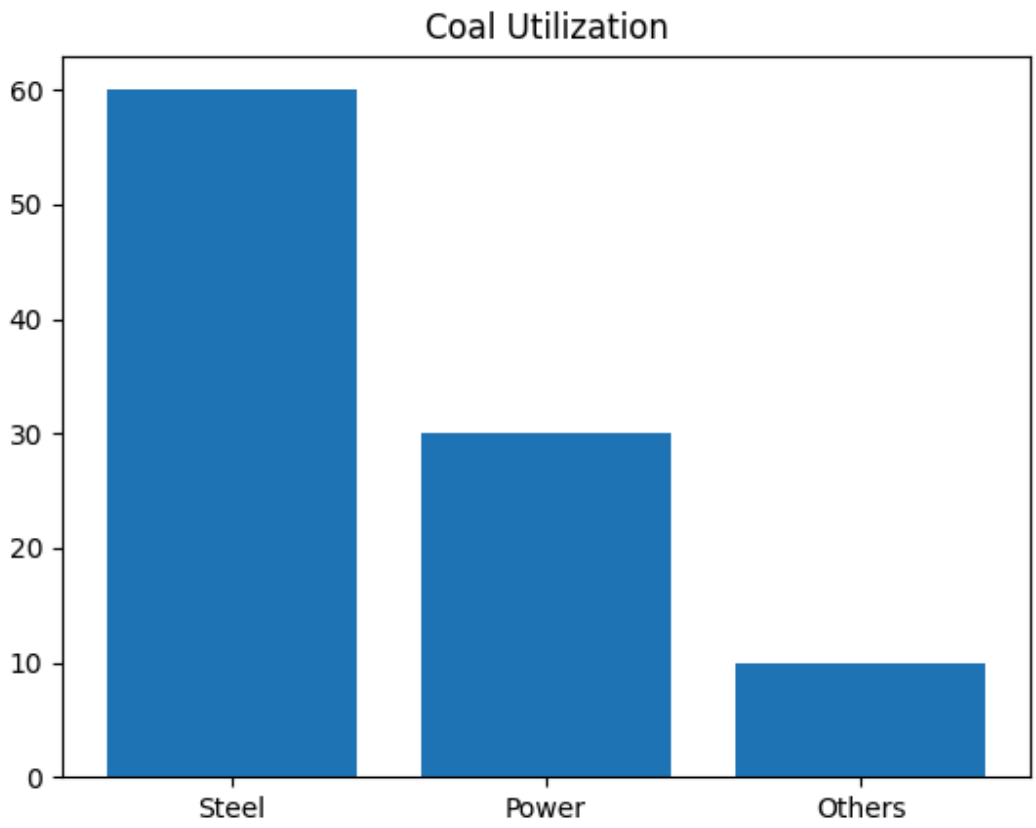
Shows breakup of mining costs into different components.
Mining and transport form the largest share.
Significance: Helps in cost control, budgeting, and improving operational efficiency.

📊 9. Coal Dispatch Mode



Represents the mode of coal transportation (rail vs road).
Rail transport dominates due to bulk handling efficiency.
Significance: Important for logistics planning and reducing transportation cost.

📊 10. Coal Utilization



Shows how coal is used across sectors like steel, power, and others.
Steel sector consumes the majority due to coking coal demand.
Significance: Highlights the industrial importance of East Bokaro coalfield.